

ZIDIO DEVELOPMENT • CLIENT PROJECT

Project FORESIGHT

Demand & Inventory Intelligence Platform

Client Account:	NorthBay Living (D2C Home & Lifestyle)
Domain:	Data Science & Supply Chain Analytics
Engagement Duration:	4-Week Client Engagement
Primary Metric:	WAPE (Weighted Absolute Percentage Error)
Lead Data Scientist:	Data Science Intern / Analyst
Repository:	codexboy0417/demand-inventory-intelligence
Status & Version:	Completed • Version 1.0

1. Executive Summary

Project FORESIGHT is an enterprise-grade demand forecasting and inventory risk intelligence engine built for **NorthBay Living**, a mid-size direct-to-consumer (D2C) home & lifestyle brand. Prior to this engagement, NorthBay Living managed inventory across ~200 active SKUs through intuition and static spreadsheets. This led to severe working capital inefficiencies: high-demand bestsellers frequently stocked out, resulting in unrecoverable revenue loss, while slow-moving products accumulated, trapping capital and forcing margin-eroding clearance markdowns.

By ingesting and processing **1,067,371 historical transaction records** spanning two full operational years, Project FORESIGHT constructs a reproducible Star-Schema data pipeline, trains multi-model machine learning demand forecasters, and connects these forward projections into an explainable **2x2 Inventory Decisioning Matrix**.

1.06M+

RAW TRANSACTIONS

72.9%

LIGHTGBM WAPE

+34.9%

WAPE IMPROVEMENT

₹1.002M

VALUE AT STAKE

Headline Business Takeaway

The LightGBM demand forecasting model achieved an out-of-sample **WAPE of 72.9%** on a 4-fold rolling-origin backtest, vastly outperforming the Seasonal-Naive baseline (107.8% WAPE). Combined with current inventory positions, the risk engine identified **₹1,000,800.85 in trapped overstock capital** across 45 clearance candidate SKUs and flagged **₹2,012.21 in immediate stockout revenue at risk** requiring purchase order replenishment.

2. Client Background & Problem Statement

NorthBay Living operates an online-only e-commerce store fulfilling customer orders from a single central warehouse. The product catalog spans furnishings, home decor, storage solutions, kitchenware, and seasonal gifts. The operations, merchandising, and finance teams faced three fundamental bottlenecks:

- **Dual Financial Loss:** Frequent stockouts on velocity items (lost revenue, customer churn) combined with excess buffer stock on dead inventory (trapped liquidity, warehouse holding costs).
- **Lack of Algorithmic Visibility:** Planning decisions were made ad-hoc in spreadsheets without statistical feature extraction or seasonality tracking.
- **Tooling Usability Gap:** Prior data science models remained trapped in code notebooks that operational and merchandising teams could not interact with or interpret.

3. Data Architecture & Data Quality Audit

The data foundation was modeled using real-world retail transactions from the **Online Retail II dataset**. To ensure scalability and clean separation of concerns, the raw records were transformed into a production **Star Schema** comprising four structured tables:

Table Name	Grain & Key Dimensions	Record Count	Core Attributes
sales_daily	Daily SKU Transactions (Fact)	532,839	date, sku_id, units_sold, revenue, unit_price, promo_flag
sku_master	Product Catalog (Dimension)	4,917	sku_id, category, subcategory, launch_date, unit_cost, list_price
calendar	Date Calendar (Dimension)	739	date, week, month, season, is_holiday, promo_event
inventory_snapshots	Current Warehouse Positions	200	date, sku_id, on_hand_units, on_order_units, lead_time_days, reorder_point

3.1 Data Cleaning Remediations

- **Order Cancellations & Returns:** Negative quantities (invoice prefix 'C') were isolated and excluded from positive sales volume to avoid demand under-counting.
- **Bad Debt & Adjustments:** Transactions with Price ≤ 0 (internal tests, accounting write-offs) were systematically purged.
- **Missing Customer IDs:** Handled guest checkout transactions by retaining sales volumes for SKU aggregation while tagging unauthenticated customer rows.
- **Catalog Categorization NLP:** Raw item descriptions lacked standard taxonomies. An automated keyword categorization parser mapped products into 5 structured categories: *Kitchen & Dining*, *Storage & Accessories*, *Home Decor & Lighting*, *Seasonal & Gifts*, and *General Furnishings*.

4. Exploratory Data Analysis (EDA) & Demand Insights

Exploratory data analysis revealed three critical operational dynamics governing NorthBay Living's sales velocity:

1. **Severe Q4 Holiday Seasonality:** Monthly revenue surges dramatically between October and December. Over **38% of annual revenue** is concentrated in Q4 due to Black Friday promotions and holiday gift purchasing. Models must explicitly incorporate 52-week seasonal lags to capture annual peaks.

2. **The 80/20 Pareto Volume Distribution:** Catalog sales are heavily skewed—the **top 15% of active SKUs generate 68% of total unit volume**. Stockouts in these high-velocity items disproportionately impact brand revenue.
3. **Catalog Dormancy & Dead Stock:** Approximately **18% of catalog SKUs exhibited zero unit sales over consecutive 60-day windows**. Holding substantial warehouse stock on these items represents locked working capital that degrades over time.

5. Feature Engineering & Leakage Prevention

Daily sales were aggregated into weekly SKU observations. To guarantee zero lookahead data leakage during backtesting and live inference, all temporal features were strictly shifted by at least one period before calculation:

Feature Group	Engineered Variables	Business Rationale
Temporal Lags	lag_1w, lag_2w, lag_4w, lag_52w	Captures immediate momentum and exact same-week annual seasonality.
Rolling Moving Statistics	rolling_mean_4w, rolling_std_4w, rolling_mean_8w	Measures short-term moving baselines and demand volatility.
Calendar & Seasonal	week, month, season, is_holiday	Provides cyclical temporal signals to tree algorithms.
Pricing & Promotions	promo_days, price_discount_ratio	Quantifies promotional discount depth and volume elasticity.

6. Machine Learning Demand Forecasting & Backtesting

In accordance with professional forecasting governance, models were evaluated using **4-fold Rolling-Origin Cross-Validation** over a 4-week test horizon. Models were measured against a **Seasonal-Naive Baseline** using Weighted Absolute Percentage Error (WAPE), Mean Absolute Percentage Error (MAPE), and Signed Forecast Bias.

Model Algorithm	Mean Out-of-Sample WAPE	WAPE vs Baseline	Model Evaluation & Selection Status
Seasonal-Naive Baseline	107.8%	<i>Reference Benchmark</i>	Baseline benchmark to clear.
XGBoost Regressor	76.3%	+31.5% Better	High accuracy; slightly slower training.
Random Forest Regressor	75.5%	+32.3% Better	Robust ensemble; higher memory footprint.
LightGBM Regressor (Winner)	72.9%	+34.9% WAPE Reduction	 Selected Production Forecaster.

7. Inventory Risk Scoring Engine & Decision Matrix

Machine learning predictions are decoupled from inventory risk rules to guarantee transparency and explainability for operations managers. The engine evaluates stock positions by comparing forward demand against current physical inventory:

7.1 Mathematical Formulation

- **Lead Time Demand (\$D_{LT}\$):** $D_{LT} = \text{Weekly Forecast} \times \left(\frac{\text{Lead Time Days}}{7.0}\right)$
- **Total Available Stock (\$S_{avail}\$):** $S_{avail} = \text{On-Hand Stock} + \text{On-Order Stock}$
- **Stockout Risk Score:** $\text{Stockout Risk} = \text{clip}\left(\frac{D_{LT} - S_{avail}}{D_{LT} + \epsilon}, 0.0, 1.0\right)$
- **Overstock Risk Score (6-Week Horizon):** $\text{Overstock Risk} = \text{clip}\left(\frac{\text{On-Hand Stock} - (\text{Weekly Forecast} \times 6)}{\text{On-Hand Stock} + \epsilon}, 0.0, 1.0\right)$
- **Financial Risk (INR):** $\text{Sales at Risk} = \text{Stockout Gap} \times \text{List Price}$ • $\text{Capital Locked} = \text{Overstock Gap} \times \text{Unit Cost}$

7.2 2x2 Decisioning Matrix Results

Decision Quadrant	SKU Count	Financial Value (INR)	Prescribed Operational Action
 Reorder Now	1 SKU	₹2,012.21 (Sales at Risk)	Issue immediate replenishment purchase order before stock depresses.
 Markdown / Clear	45 SKUs	₹1,000,800.85 (Locked Capital)	Initiate promotional discounting campaign to liquidate excess stock.
 Healthy	154 SKUs	₹0.00 (Optimal Balance)	Inventory matches demand velocity; maintain current replenishment cycles.
 Watch / Volatile	0 SKUs	₹0.00	Erratic demand fluctuations; flag for manual merchandising review.

8. Productized Interfaces & Deployment Architecture

The analytics stack is packaged into two stakeholder-facing interfaces:

- **7-Page Streamlit Planning Dashboard (app/main.py):**
 - *Home Page:* Executive summary banner, active SKU counters, architecture overview.
 - *Sales Analytics:* Monthly revenue trends, Pareto 80/20 top-mover bar charts, category pie charts.
 - *Forecast:* Backtest comparison table and interactive SKU forward forecast generator with 80% confidence

bands.

- *Inventory Dashboard*: On-hand vs on-order stacked bar charts and supplier lead-time breakdowns.
- *Risk Dashboard*: Dynamic 2x2 Plotly scatter plot (bubble size = ₹ value at stake) with quadrant filter tabs.
- *Product Details*: SKU deep-dive searching, margin economics, and prescribed operational actions.
- *Executive Summary Dashboard*: High-level ROI metrics, Top 5 Reorder Priorities, Top 5 Markdown Candidates.
- **FastAPI Scoring Microservice (service/api.py)**:
 - GET /forecast/{sku_id}: Real-time single-SKU forward demand and risk score lookup.
 - POST /score-batch: Batch scoring microservice for ERP/WMS system integrations.

9. Strategic Business Recommendations & Next Steps

Based on the quantitative findings of Project FORESIGHT, NorthBay Living leadership should execute the following 3-step action roadmap:

1. Immediate Reorder Execution (Protect Revenue)

Execute purchase order replenishment for **SKU 21213** (Storage & Accessories) to protect against imminent stockout revenue losses during upcoming lead-time fulfillment.

2. Capital Liberation Clearance Campaign (Recover ₹1.00M Liquidity)

Launch structured markdown promotions across the **Top 3 Overstock SKUs (23843, 22086, 37410)**, liberating ₹1,000,800.85 in trapped working capital to reinvest into high-velocity inventory lines.

3. Monthly Automated Pipeline Cadence

Integrate the automated pipeline script (`python src/risk.py lightgbm`) into a monthly batch cron schedule to continuously refresh SKU forecasts as new monthly inventory and sales extracts arrive.